

Praktikum

Design of Wind Farms

Lecture 2:
Development of a Wind Farm Project

Hadi Hoghooghi
Technische Universität München



Syllabus

No.	Week	Date (Wed)	9.30–10.15	10.30–11.15	11.30–12.15
1	14	15/04/26	Overview of Course Lecture 1: Background and Motivations	Lecture 2: Development of a Wind Farm Project	Lecture 3: Wake Behavior, Wind Farm Control Assignment 1: Basics and Recapitulation
2	15	22/04/26	Lecture 4: Wake Models, FLORIS	Assignment 1: Solution Assignment 2: Jensen's Wake Model and Wake Superposition in Wind Farms	Tutorial 1: FLORIS Installation and Setup
3	16	29/04/26	Lecture 5_1: LandBOSSE	Tutorial 2: LANDBOSSE Installation and Setup	Assignment 2: Solution Assignment 3: Wind Farm Analysis Using FLORIS
5	17	06/05/26	Lecture 5_2: Financial Analysis	Assignment 3: Solution	Assignment 4: Wind Farm Cost Analysis Using FLORIS + LandBOSSE)
6	18	13/05/26	Lecture 6: Layout Optimization	Assignment 4: Solution	Assignment 5: Layout Optimization
7	19	20/05/26	Assignment 5: Solution		Final group assignment: Introduction, Finalizing Groups, Q&A
8	20	27/05/26	Final group assignment: Design Process, Q&A, Writing Report		
9	21	03/06/26	No Class		
10	22	10/06/26	Final group assignment: Design Process, Q&A, Writing Report		
11	28	08/07/26	No Class Deadline: Electronic report submission until Tuesday, 07/07/2026, 23:59h		
12	29	15/07/26	Final presentation – 8:30h (Room 2701M)		



Agenda

- Development of a Wind Farm Project
- Offshore Wind Farms



Main Elements of The Development of A Wind Farm Project

Three main elements

- Technical and commercial issues
- Environmental considerations
- Dialogue and consultation

Wind farm development phases:

- Initial site selection
- Project feasibility assessment
- Preparation and submission of the planning (permitting) application
- Construction
- Operation
- Decommissioning and land reinstatement



Initial Site Selection

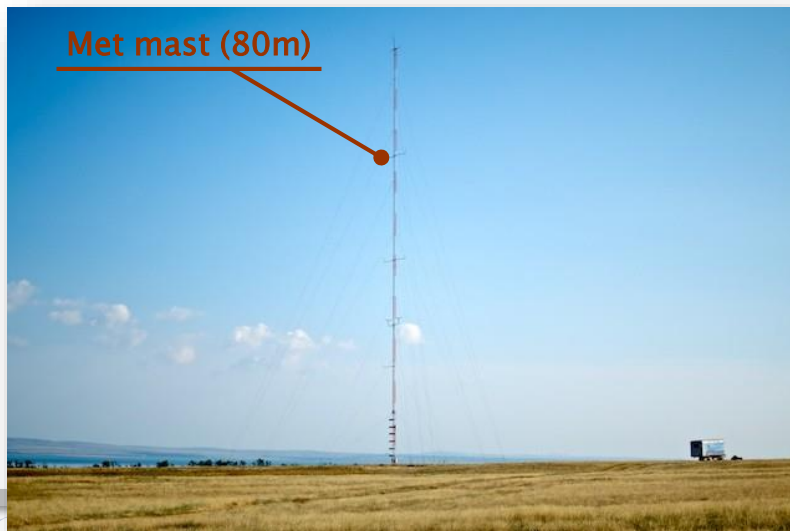


Wind Measurement Equipment

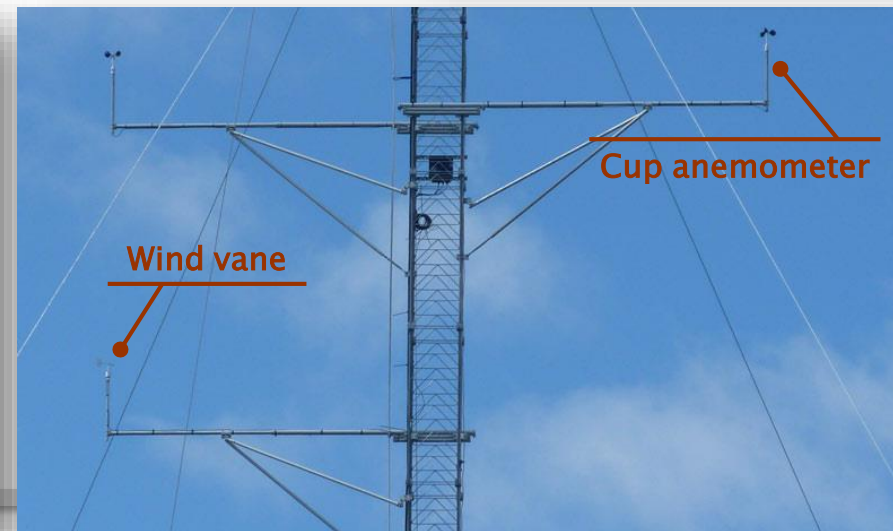
- For **site assessment**:
 - **Meteorological masts** (cup anemometers and wind vanes, sensors for pressure, temperature, humidity, ...)
 - **SODAR** (Sonic Detecting And Ranging)
 - **LiDAR** (Light Detection And Ranging)
- **Wind Atlas** for greater region
(not for detailed site examination)
- Wind measurements also used in **operation** and **performance evaluation**



(source: www.offshorewind.biz)



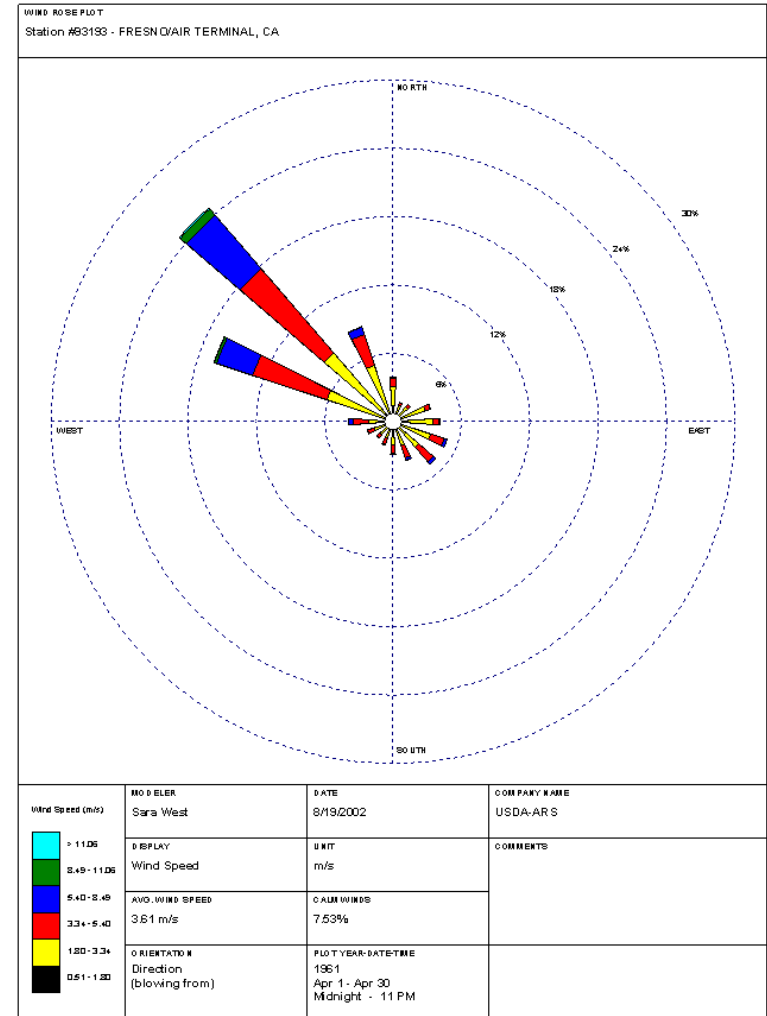
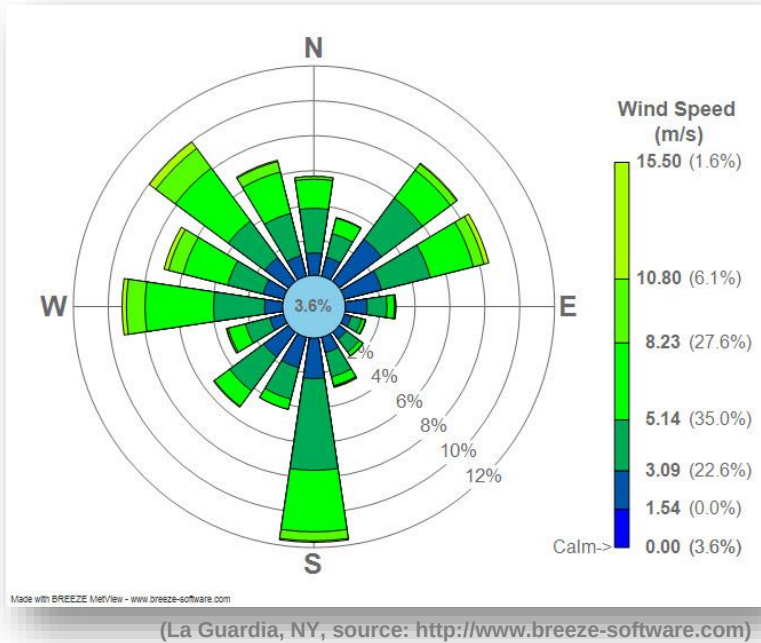
(source: www.wind-pgc.com)



(source: www.sgurrenergy.com)

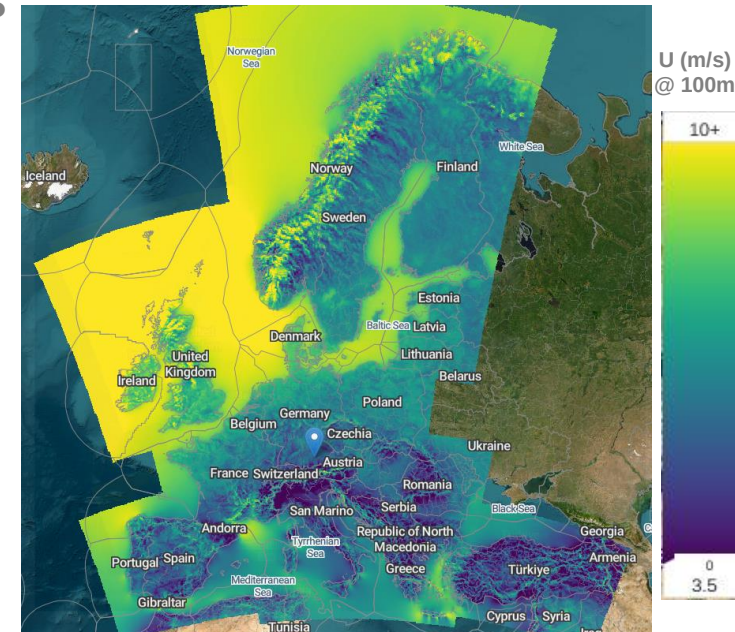
Wind Direction Distribution

- **Wind rose** diagram (speed, direction and frequency)
- Important for farm siting and layout optimization



New European Wind Atlas

- Wind roses are created using **time-series** (long-term records) of anemometer data (wind speed and direction)
- No local data available → wind atlases for resource estimate
- Open-source for Europe: **New European Wind Atlas (NEWA)**, available via <https://map.neweuropeanwindatlas.eu/>
- Base: 30 years of mesoscale simulations (resolution: 3 km, 30 min)
- Downscaling with linearized microscale model (50 m grid)
- Available information: wind resource, site suitability, wind variability, wind power predictability, etc. at different heights



New European Wind Atlas

- MESO time-series can be downloaded for period 2005 – 2018
- NetCDF-File (**.nc**) can be easily processed with MATLAB functions **ncinfo** and **ncread**
- Processing with FLORIS easiest by creating **.csv** file with wind resource information
- **Example .nc to .csv conversion using MATLAB:**
WS10 = ncread("Kastrup.nc","WS10");
WD10 = ncread("Kastrup.nc","WD10");
KastrupTS = table(WS10,WD10);
writetable(KastrupTS,"KastrupTimeSeries.csv");
- Shear exponent can be calculated using data from different heights
- Turbulent Kinetic Energy (TKE) and Turbulence Intensity (TI):

$$TKE = \frac{3}{2} (TI * U_{mean})^2$$

Influence of Wind Data Binning on Simulation Results

- Discretization effects on wind rose and simulation results:
 - The resolution of wind speed and wind direction bins affects the shape of the wind rose.
 - When reading data with software like FLORIS, coarse discretization can alter predicted flow patterns and power output.
 - Typical practice:
 - Wind direction: bins of up to 5°
 - Wind speed: bins of up to 1 m/s
 - Finer discretization improves accuracy but increases computational cost.

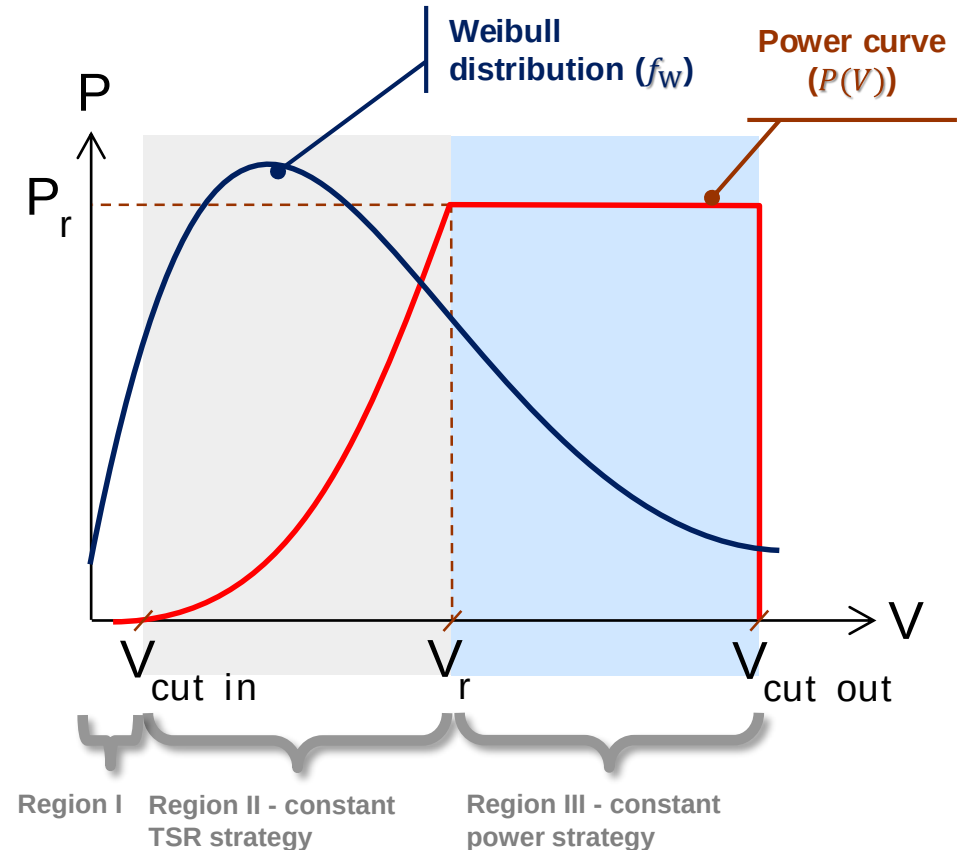


Weibull Distribution to Compute AEP

Annual energy production:

$$AEP = Y \int_{V_{in}}^{V_{out}} P(V) f_W dV$$

with turbine power curve $P(V)$
and $Y = 8760$ h/year



Weibull Distribution

Probability distribution function, used to describe the probability of occurrence of a certain wind speed

- **Cumulative** probability function (i.e. probability that $V < V_0$):

$$F_W(V_0) = \int_0^{V_0} f_W(V') dV' = 1 - \exp\left(-\left(\frac{V_0}{C}\right)^k\right)$$

- Probability **density** function:

$$f_W(V_0) = k \frac{V_0^{k-1}}{C^k} \exp\left(-\left(\frac{V_0}{C}\right)^k\right)$$

with

k

$$C = \frac{V_{ave}}{\Gamma(1+1/k)}$$

V_0

V_{ave}

$\Gamma(\cdot)$

shape parameter of the Weibull function,
scale parameter of the Weibull function [m/s],
 wind speed [m/s],
 average wind speed [m/s] and
 gamma function ($\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$)

- For $k=2$ (typical value for many locations): **Rayleigh Distribution**



Capacity Factor

Power plants do not work at full power at all times

Wind turbines:

- Environmental variability (wind and air density)
- Availability (fraction of time WT is able to produce electricity, i.e. not down for scheduled or unscheduled maintenance) (typically > 95%)

Capacity factor:

(produced energy in given period) = **(capacity factor)** * (theoretical max energy)

(theoretical max energy) = (nameplate power) * (period length)

▼ Wind (source: IEA)

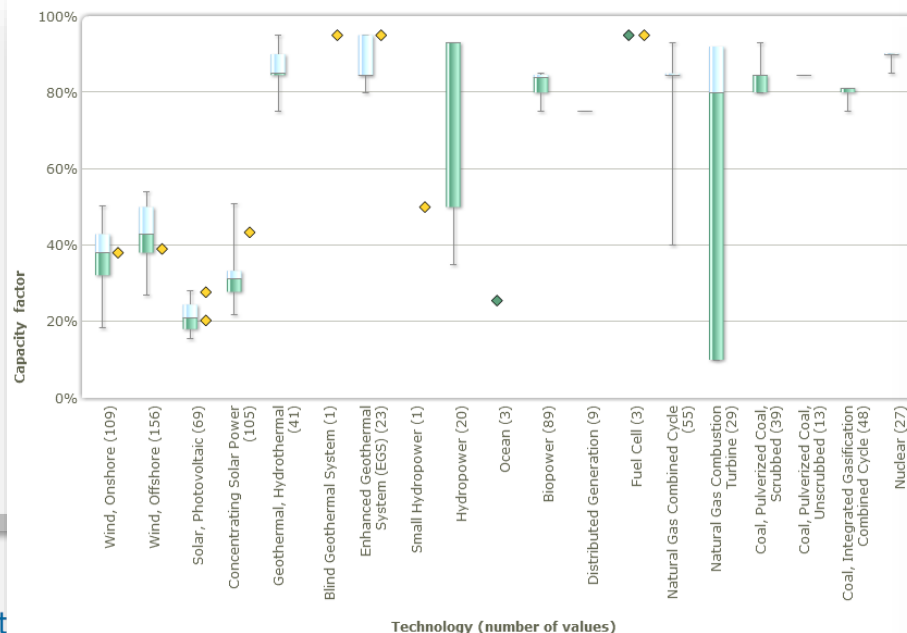
Capacity factor:

$$CF = \frac{AEP}{Y \cdot P_r}$$

$$0.2 < CF < 0.6$$

Capacity factors for various technologies

(source: en.openei.org)



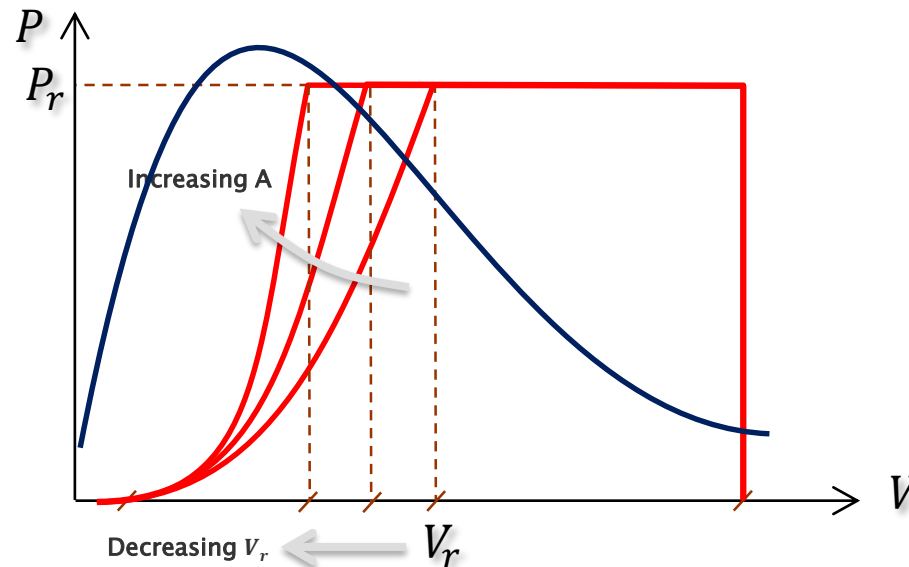
Country	2011 Average capacity factor (%)	2012 Average capacity factor (%)
Australia	---	35.0
Austria	---	30.0
Canada	31.0	31.0
China	---	18.4
Denmark	28.4	22.6
Finland	28.0	24.0
Germany	19.0	---
Greece	---	---
Ireland	31.6	28.4
Italy	18.0	---
Japan	19.0	19.9
Korea	---	---
México	30.0	30.0
Netherlands	---	On land 20.0 Offshore 39.5
Norway	31.3	31.2
Portugal	26.0	28.0
Spain	---	24.1
Sweden	---	26.0
Switzerland	20.0	<20.0
United Kingdom	On land 27.4 Offshore 36.7	On land 27.4 Offshore 36.7
United States	33.0	33.0



Specific Power

Rated power: $P_r = \frac{1}{2} \rho A V_r^3 C_{P_{\max}}$ Rated wind speed $V_r = \sqrt[3]{\frac{P_r/A}{\frac{1}{2} \rho C_{P_{\max}}}}$

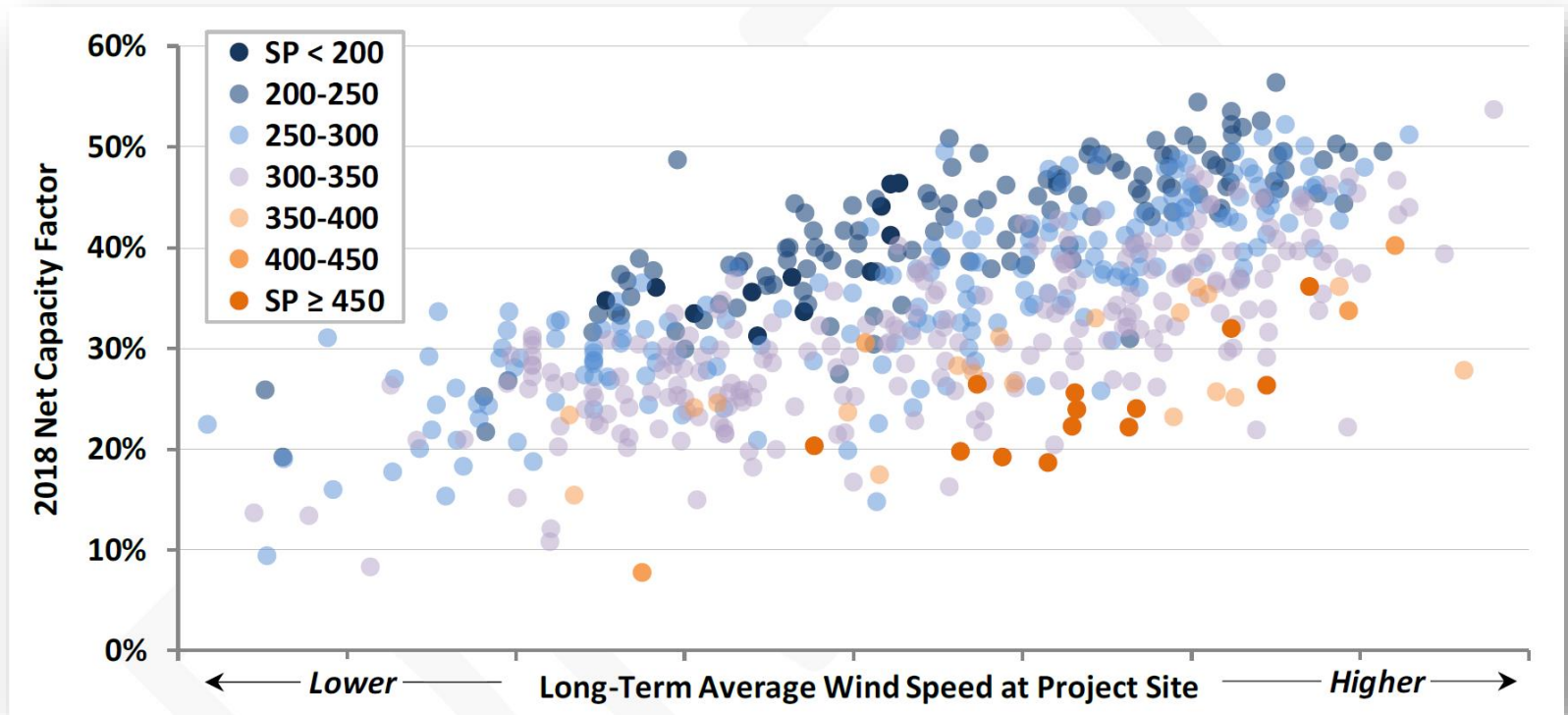
- $C_{P_{\max}}$ limited by physics
- To decrease $V_r \Rightarrow$ reduce specific power (power loading) P_r/A



More time spent in region III at full power, **increased capacity factor**



Correlation between Capacity Factor and Specific Power



(Source: M. Bolinger et al., 2019)



Levelized Cost of Electricity (LCOE)

Levelized cost of electricity (LCOE) is defined as the price at which the generated electricity should be sold for the system to break even at the end of its lifetime.

A simple formulation for called sLCOE can be written as follows:

$$sLCOE = \frac{(Rated\ Capacity \times Investment\ Cost \times CRF) + (O\&M\ Cost \times AEP + \dots)}{Rated\ Capacity \times Capacity\ Factor \times 8760\ h}$$

Capital Recovery Factor: A capital recovery factor is the ratio of a constant annuity to the present value of receiving that annuity for a given length of time.

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1}$$

i = discount rate, n = project lifetime (e.g. 20 years)



Vertical Wind Profile (Shear)

Wind speed increase with height within the atmospheric boundary layer
Strong effect on **power production** and **fatigue loading**

Profile models:

- **Logarithmic** (derived from BL theory):

$$U(z) = U(z_r) \frac{\ln(z/z_0)}{\ln(z_r/z_0)}$$

- **Power law** (flat plate, but also empirical):

$$U(z) = U(z_r) \left(\frac{z}{z_r} \right)^\alpha$$

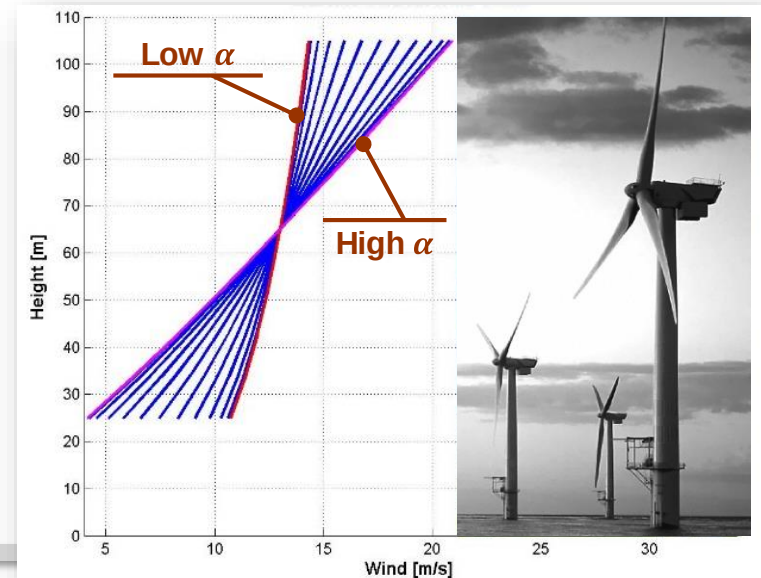
z_r = reference height

z_0 = surface roughness length (crops, trees, buildings, waves, ...)

α = wind shear (power law) exponent, depends on stability, terrain features, wind speed, ... (typically 0.2 on-shore, 0.14 off-shore)

Normal Wind Profile (NWP): $z_r = z_{\text{hub}}$; $\alpha = 0.2$

- Actual wind can be seen as **superposition of vertical shear and turbulent fluctuations**



IEC Wind Classes

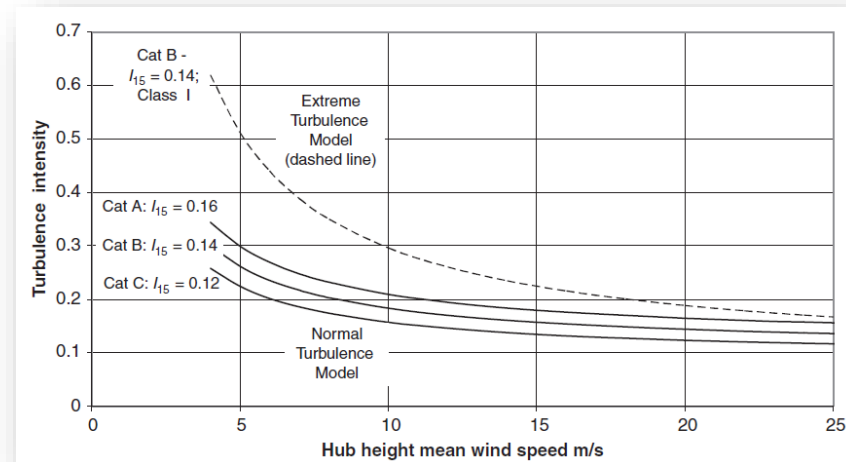
Wind model used for **design and certification** of wind turbines

IEC 61400-1 defines:

- Annual **average wind speed** (V_{hub})
- **Reference wind speed** or 10 min. mean with 50-year return period (V_{ref})
- **Turbulence intensity** (I_{ref} at hub height and 15 m/s)
- Extreme 50-year gust ($1.4 V_{\text{ref}}$)
- ...

IEC 61400: International Standard published by the International Electrotechnical Commission

For turbine certification, a large number of **design load cases** (DLCs) for different wind conditions are simulated



(turbulence models, source: Wind Energy Handbook, 2nd Ed.)

Class	I	II	III	S
V _{hub} (m/s)	10	8.5	7.5	Values specified by designer
V _{ref} (m/s)	50	42.5	37.5	
A I _{ref}	0.16			
B I _{ref}	0.14			
C I _{ref}	0.12			

Initial Site Selection

Typical separation distances for medium sized wind turbines

Feature	Typical separation distances
Residential property	350–500 m
Motorway, trunk road, railway	Tip of turbine +10%
Public right-of-way (e.g. footpath)	50 m – tip of turbine, depending on use of right-of-way
Electric overhead line	Tip of turbine +10%
Turbines	5–6 rotor diameters in prevailing wind direction 3–4 rotor diameters across prevailing wind direction

Project Feasibility Assessment

Once a potential site has been identified then more detailed, and expensive, investigations are required in order to confirm the feasibility of the project.

It is not generally considered acceptable in complex terrain to rely on the estimates of wind speed made during the initial site selection but to use the **Measure Correlate-Predict (MCP)** technique to establish a prediction of the long-term wind resource.

MCP requires the installation of a **mast** at the wind farm site on which are mounted anemometers and a wind vane.

- If possible one anemometer is mounted at the **hub height** with others lower to allow **wind shear** to be measured.
- Measurements are made over at least a **six-month** period.

Micrositing and Site Investigations

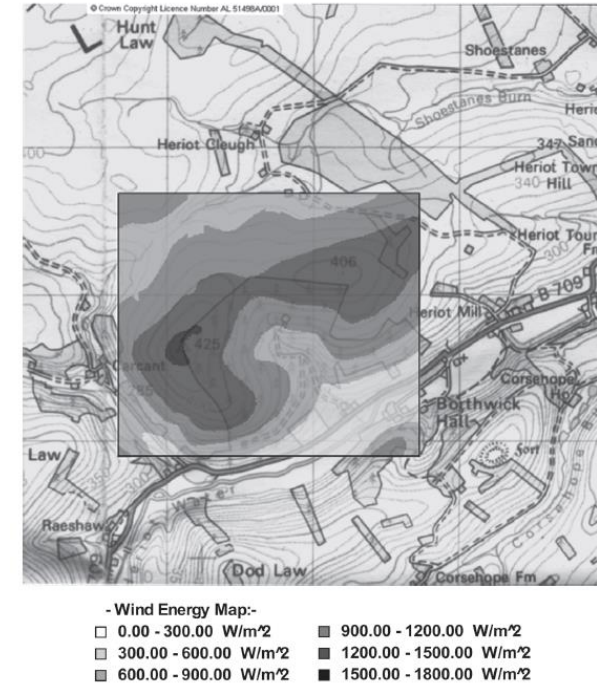
In addition to MCP, **design software package** to investigate the performance of a number of potential turbine layouts.

Two common models:

- **Wind Atlas Analysis** and
 - **Application Program (WAsP) and MSMicro/3**
- Calculate **wind flows** over a site
 - Other constraints such as **turbine separation, terrain slope, wind turbine noise, radar and land ownership boundaries** may also be applied
 - **3D virtual reality techniques**

Investigation:

- Careful assessment of existing land use
- Ground conditions (turbine foundations, access roads and construction areas)



Source: Wind Energy Handbook



Public Consultation and Preparation; Submission of the Planning Application

Informal public consultation:

local community, organisations, environmental societies and wildlife trusts

The purpose of a wind farm Environmental Statement may be summarised as to:

- Physical characteristics of the wind turbines and their land-use requirements
- Environmental character of the proposed site and the surrounding area
- Environmental impact of the wind farm
- Measures which will be taken to mitigate any adverse impact
- The need for the wind farm and provide details to allow the planning

Topics covered in the Environmental Statement:

- Policy framework
- Site selection
- Designated areas
- Visual and landscape assessment
- Noise assessment
- Ecological assessment
- Archaeological and historical assessment
- Hydrological assessment
- Interference with telecommunication systems
- Aircraft safety and interference with radars
- Safety
- Traffic management and access construction
- Electrical connection
- Economic effects on the local economy
- Decommissioning
- Mitigating measures
- Non-technical summary



Landscape and Visual Impact Assessment

- Landscape character assessment (including landscape policy and designation)
- Design and mitigation
- Assessment of impacts (including visibility and viewpoint analysis)
- Shadow flicker



Wind farm of six 660 kW turbines in flat terrain

Source: Wind Energy Handbook



Landscape Character Assessment

- The fundamental step in minimising the **visual impact** of a wind farm is to identify an appropriate site and ensure that the proposed development is in **harmony with the location**.



Wind farm of 600 kW turbines, Tarifa, Spain.



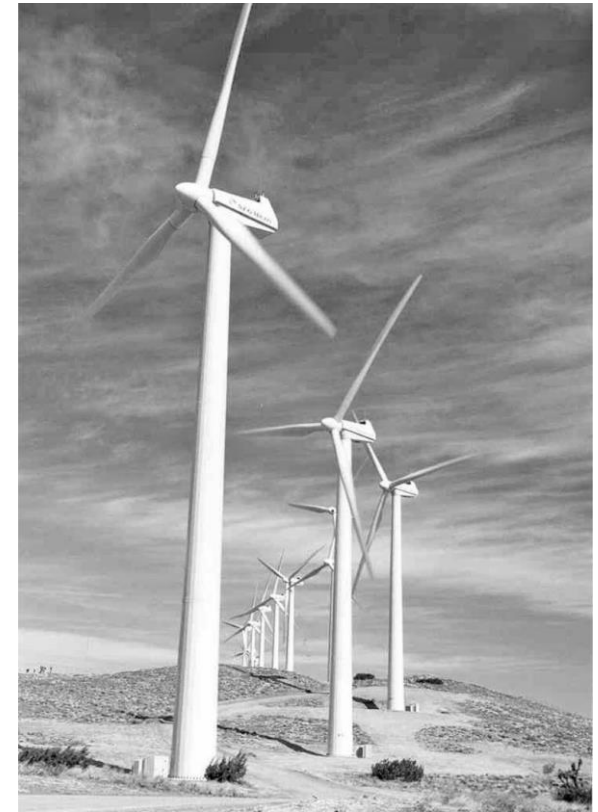
Wind farm of 700 kW wind turbines located along coast.

Source: Wind
Energy Handbook



Design and Mitigation

- Lower **speed of rotation** is more relaxing to the eye
- **Tower heights** and **tower materials**
- **Turbine color**
- The **layout** and design of the wind farm
- Associated **ancillary structures** required (substation)
- **Roads**
- ...



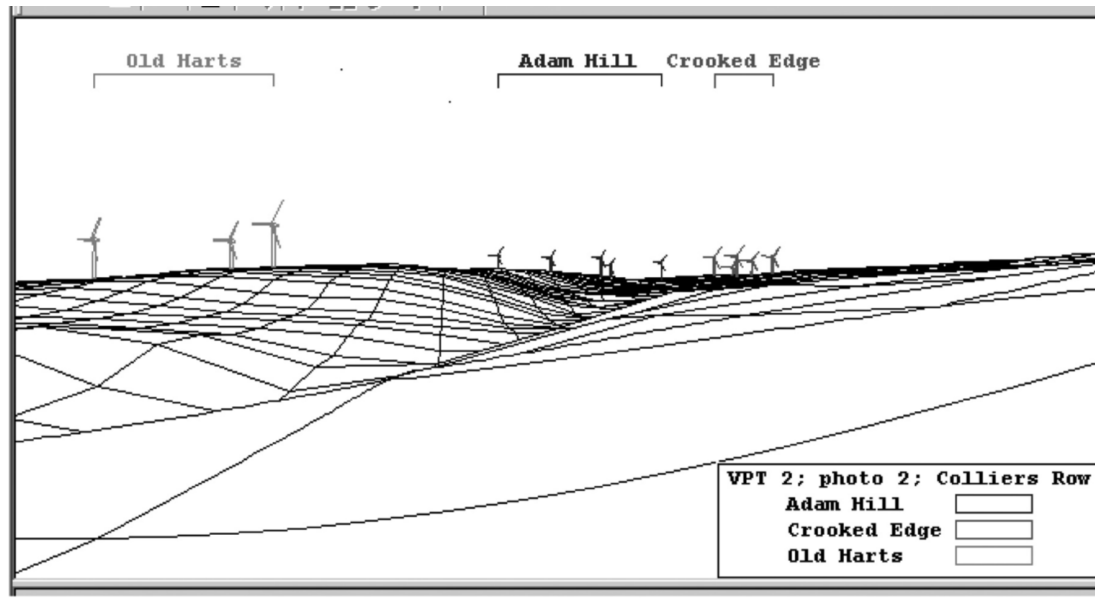
Cameron Ridge in California, USA

Source: Wind
Energy Handbook



Assessment of Impact

- A major part of the Environmental Statement is the **assessment of landscape and visual impact**
- **Viewpoint analysis**
 - Sensitivity of the viewpoint (residential/industrial)
 - Magnitude of the change of view



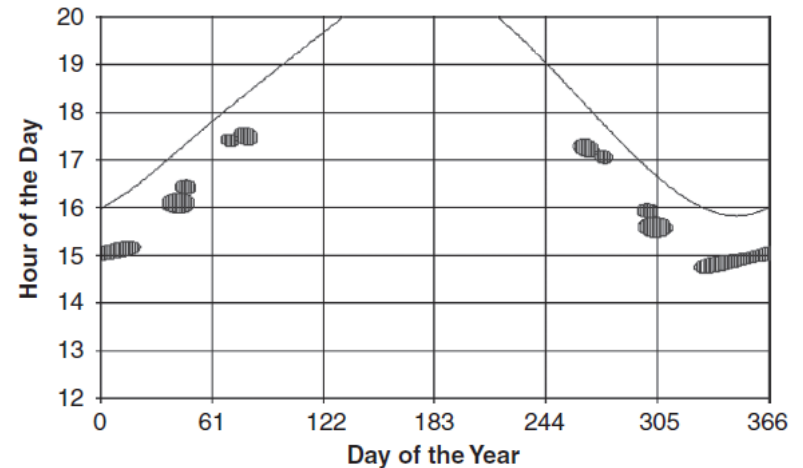
Example of wire-frame showing visibility of three wind farms.

Source: Wind Energy Handbook



Shadow Flicker

- **Shadow flicker** is the term used to describe the stroboscopic effect of the shadows cast by rotating blades of wind turbines when the sun is behind them.
 - The frequencies that can cause disturbance are between **2.5–20 Hz**
 - The effect on the **human eye-brain** is the same as that caused by changes in intensity of an incandescent electric light due to variations in network voltage from a wind turbine
 - Main concern is variations in light at frequencies of **2.5–3 Hz**
 - Large modern three-bladed wind turbines will rotate at less than 30–35 rpm, giving **blade passing frequencies of 1.5–1.75 Hz**, which is below the critical frequency of 2.5 Hz.



Example of shadow flicker prediction –
continuous line shows time of sunset



Noise

- Wind farm Environmental Statement should include:
 - Predicted **noise levels** at specific properties close to the wind farm over the most critical range of wind speeds
 - Measured **background noise levels** at the properties at these wind speeds
 - A **scale map** showing the proposed wind turbines, the prevailing wind conditions and nearby existing developments
 - Results of independent measurements of **noise emission** from the proposed wind turbines including the **sound power and narrow band frequency spectrum**
- **IEC 61400–11 (2006)** describes how tests to determine the noise emissions from a wind turbine may be conducted
- **IEC 61400–14 (2005)** details how the results of these tests should be declared



Wind Turbine Noise

- Wind turbine noise:
 - **Mechanical** (rotating machinery in the nacelle, particularly the gearbox and generator, cooling fans, auxiliary equipment, yaw system)
 - **Aerodynamic** (low frequency noise (dominated by 3P), inflow turbulence noise, airfoil self noise)
- Airfoil self noise:
 - Trailing edge noise (750–2000 Hz)
 - Tip noise
 - Stall effects
 - Blunt trailing edge noise
 - Surface imperfections

sound power level:

$$LW \approx (50 \text{ to } 60) \log V_{\text{tip}}$$

Element	Sound power level dB(A)	Air-borne or structure-borne
Gearbox	97.2	Structure-borne
Gearbox	84.2	Air-borne
Generator	84.2	Air-borne
Hub	89.2	Structure-borne
Blades	91.2	Structure-borne
Tower	71.2	Structure-borne
Auxiliaries	76.2	Air-borne



Measurement, Prediction and Assessment of Wind Farm Noise

- The sound power level of a wind turbine is normally determined by field experiments.
- The measurements are taken at a distance, R_0 , from the base of the tower:

Country	Commercial	Mixed	Residential	Rural
Germany				
Day	65	60	55	50
Night	50	45	40	35
Netherlands				
Day		50	45	40
Night		40	35	30
Denmark			40	45

$$R_0 = H + \frac{D}{2}$$

- Sound pressure level of the wind turbine:

$$L_P = 10 \log_{10} \left(10^{\frac{L_{P+N}}{10}} - 10^{\frac{L_N}{10}} \right)$$

- L_P is the sound pressure level of the wind turbine.
- L_{P+N} is the sound pressure level of the wind turbine and the background sound.
- L_N is the sound pressure level of the background.

- A-weighted sound power level of the turbine:

$$L_W = L_{PAeq} + 10 \log_{10} (4\pi R_i^2) - 6$$

- R_i is the slant distance from the microphone to the wind turbine hub



Electromagnetic Interference and Aviation Radar

- Wind turbines have the potential to **interfere with electromagnetic signals** that form part of a wide range of modern communication systems and so their siting requires careful assessment in respect of **Electromagnetic Interference (EMI)**.
- Electromagnetic properties of wind turbine rotors will be influenced by:
 - Rotor diameter and rotational speed
 - Rotor surface area, planform and blade orientation including yaw angle
 - Hub height
 - Structural blade materials and surface finish
 - Hub construction
 - Surface contamination (including rain and ice)
 - Internal metallic components including lightning protection
- The main impacts of a wind farm on radar are:
 - Masking
 - Radar Clutter
 - Scattering

More details: Wind Energy Handbook



Ecological Assessment

- Categories of effects:
 - Immediate damage to **wildlife habitats** during construction
 - Direct effects on **individual species** during operation
 - Longer term changes to **wildlife habitats** as a consequence of construction or because of changed land use management practices
- Impact on birds:
 - Collision
 - Displacement due to disturbance
 - Barrier effect
 - Habitat change and loss



Off-Shore Wind

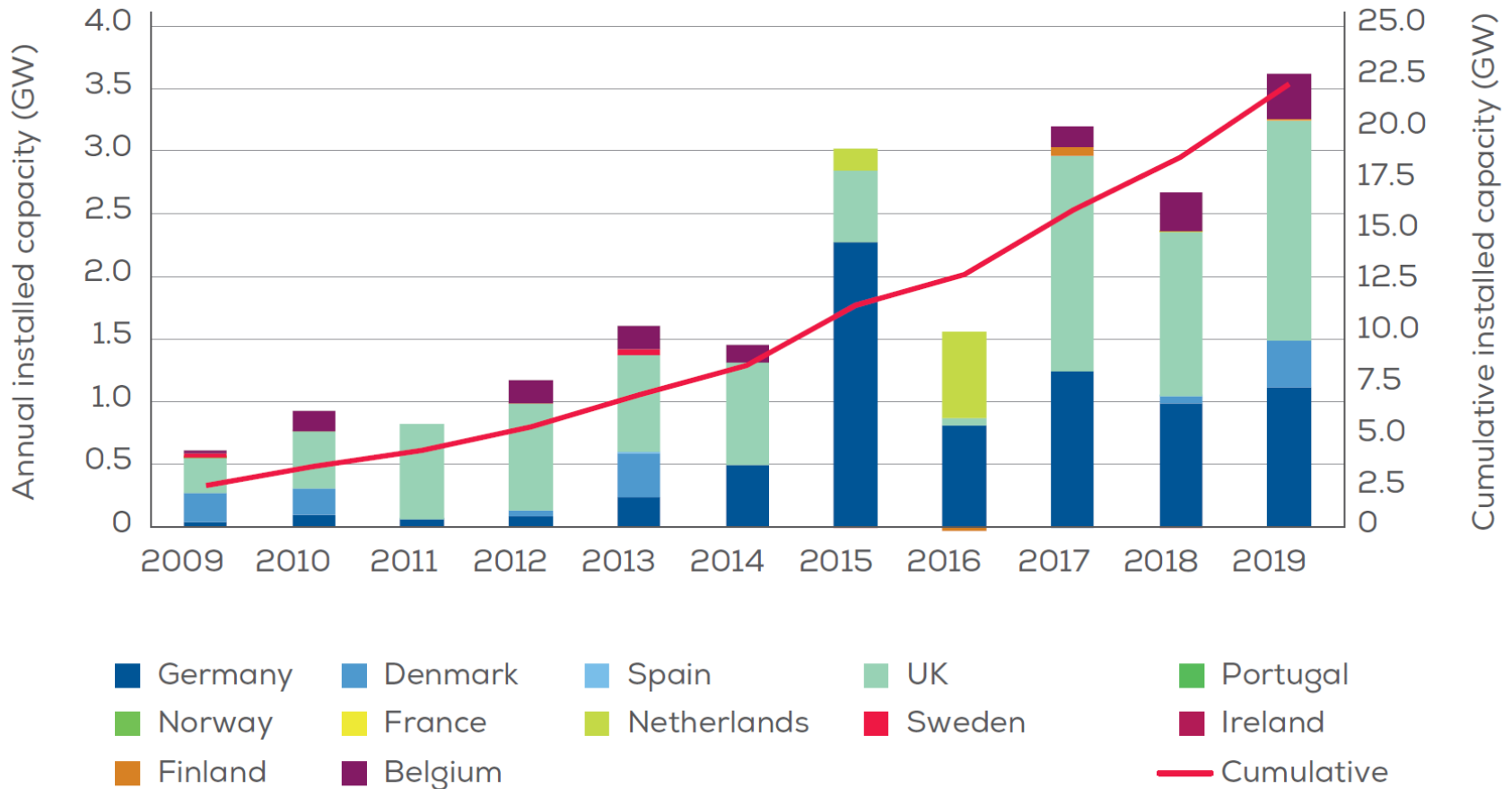
Reasons for going off-shore:

- Huge available resources
- Improved social acceptability
- Lower environmental impacts (e.g. noise)
- Scale / logistics
- Potential for reduced LCOE



Off-Shore Wind

Annual offshore wind installations by country (left axis) and cumulative capacity (right axis)

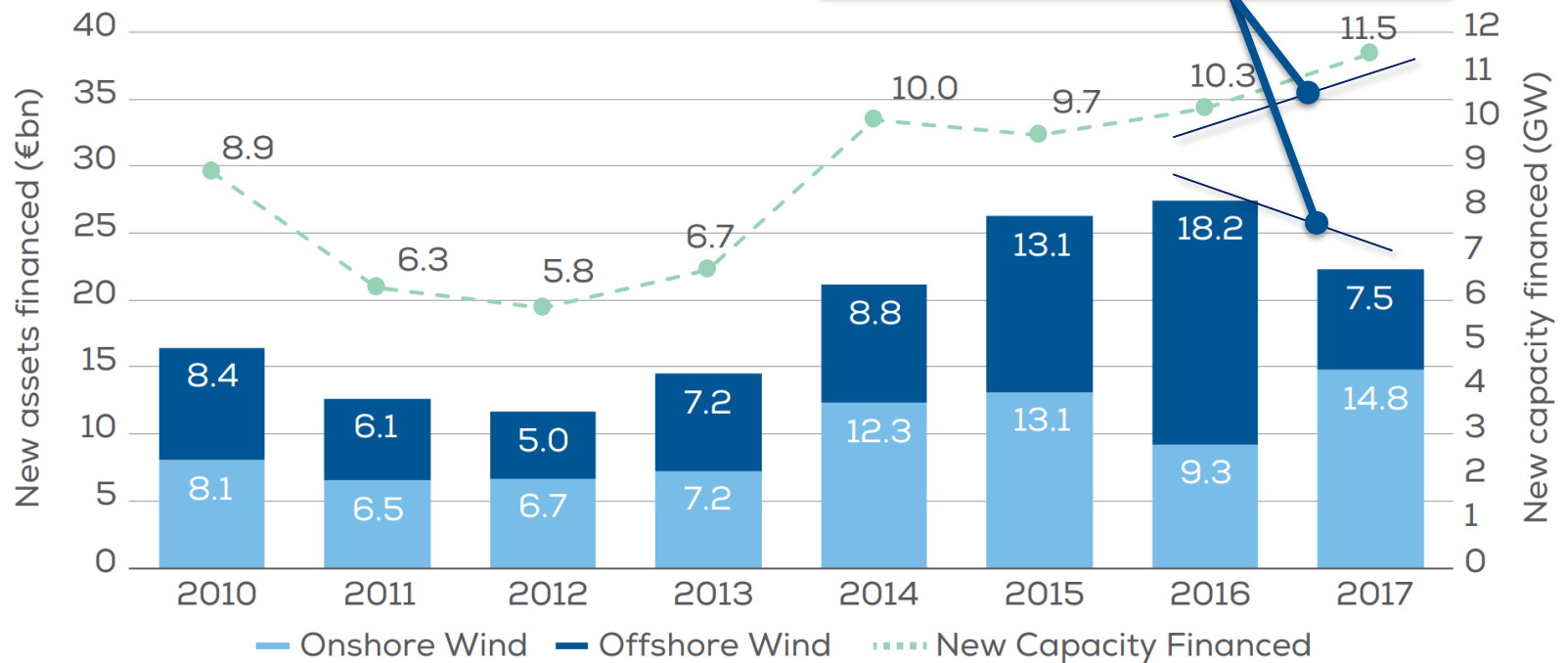


Source: WindEurope



Off-Shore Wind

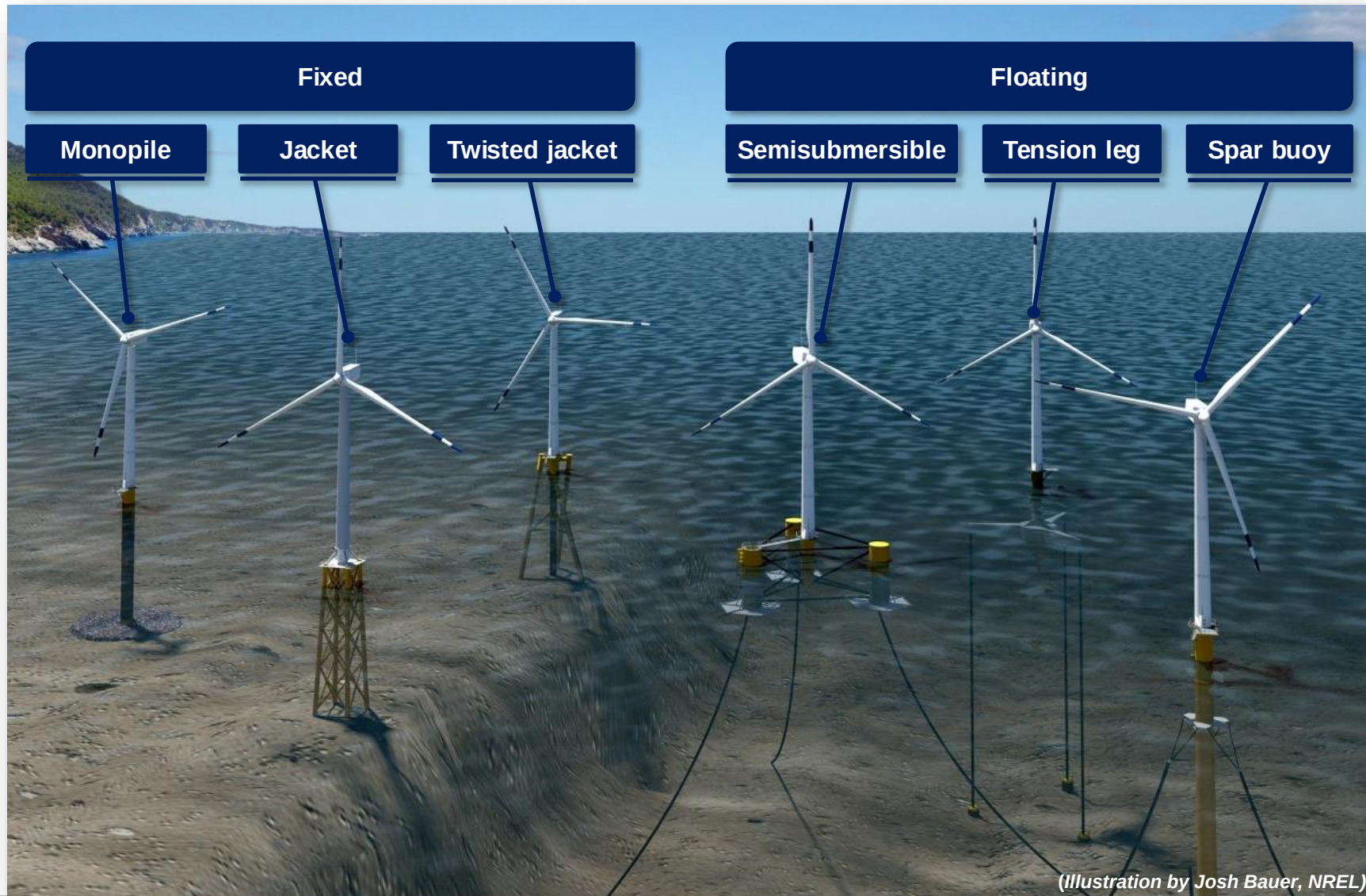
New asset finance in wind energy 2010 – 2017⁹



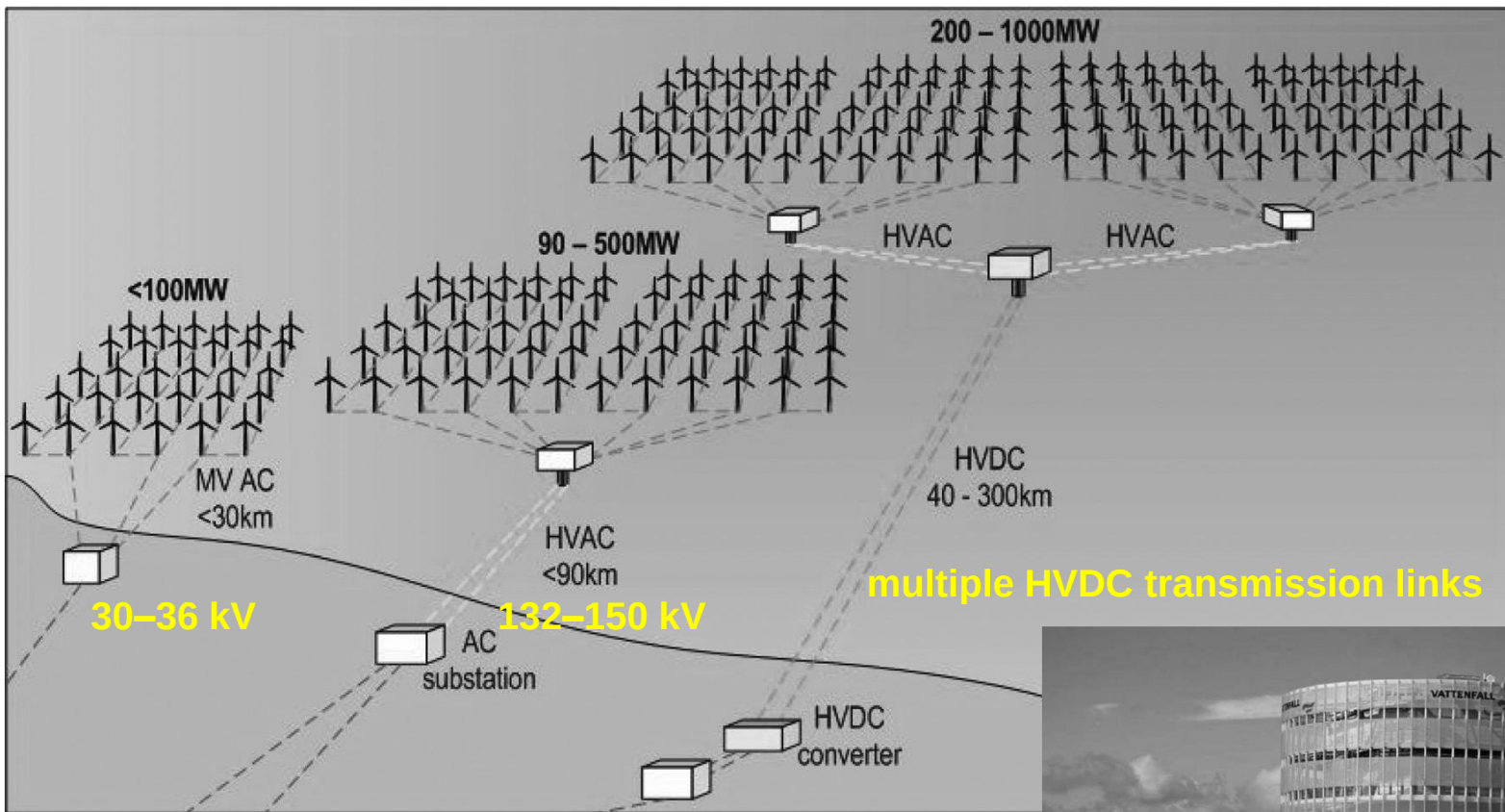
Source: WindEurope



Off-Shore Foundations



Offshore Power Collection and Transmission



- Key parameter for **layout optimization**

Offshore sub-station



More Detailed Information on Wind Farm

- 1) Tony Burton, Nick Jenkins, David Sharpe, Ervin Bossanyi, GL Garrad Hassan. Wind Energy Handbook, Second Edition, WILEY, 2011.

